

Poisson Regression and Negative Binomial Regression

Poisson Regression

Bikeshare data set contains the hourly and daily count of rental bikes between years 2011 and 2012 in Capital bikeshare system, along with weather and seasonal information. Please use “help(Bikeshare)” command for more information on the data set.

Let's build a regression model to predict the number of bikers by using the data.

```
require(ggplot2)
```

```
## Loading required package: ggplot2
```

```
require(sandwich)
```

```
## Loading required package: sandwich
```

```
require(msm)
```

```
## Loading required package: msm
```

```
#library(faraway)
```

```
library(MASS)
```

```
library(ISLR2)
```

```
##
```

```
## Attaching package: 'ISLR2'
```

```
## The following object is masked from 'package:MASS':
```

```
##
```

```
## Boston
```

```
str(Bikeshare)
```

```
## 'data.frame': 8645 obs. of 15 variables:
## $ season : num 1 1 1 1 1 1 1 1 1 1 ...
## $ mnth : Factor w/ 12 levels "Jan","Feb","March",...: 1 1 1 1 1 1 1 1 1 1 ...
## $ day : num 1 1 1 1 1 1 1 1 1 1 ...
## $ hr : Factor w/ 24 levels "0","1","2","3",...: 1 2 3 4 5 6 7 8 9 10 ...
## $ holiday : num 0 0 0 0 0 0 0 0 0 0 ...
## $ weekday : num 6 6 6 6 6 6 6 6 6 6 ...
## $ workingday: num 0 0 0 0 0 0 0 0 0 0 ...
## $ weathersit: Factor w/ 4 levels "clear","cloudy/misty",...: 1 1 1 1 1 2 1 1 1 1 ...
## $ temp : num 0.24 0.22 0.22 0.24 0.24 0.24 0.22 0.2 0.24 0.32 ...
## $ atemp : num 0.288 0.273 0.273 0.273 0.288 0.288 ...
## $ hum : num 0.81 0.8 0.8 0.75 0.75 0.75 0.8 0.86 0.75 0.76 ...
## $ windspeed : num 0 0 0 0 0 0.0896 0 0 0 0 ...
## $ casual : num 3 8 5 3 0 0 2 1 1 8 ...
## $ registered: num 13 32 27 10 1 1 0 2 7 6 ...
## $ bikers : num 16 40 32 13 1 1 2 3 8 14 ...
```

```

dim(Bikeshare)

## [1] 8645 15

names(Bikeshare)

## [1] "season"      "mnth"        "day"         "hr"          "holiday"
## [6] "weekday"     "workingday"  "weathersit"   "temp"        "atemp"
## [11] "hum"         "windspeed"   "casual"      "registered"  "bikers"

modp<- lm(
  bikers ~ mnth + hr + workingday + temp + weathersit,
  data = Bikeshare
)
summary(modp)

##
## Call:
## lm(formula = bikers ~ mnth + hr + workingday + temp + weathersit,
##     data = Bikeshare)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -299.00  -45.70   -6.23   41.08  425.29
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -68.632     5.307  -12.932 < 2e-16 ***
## mnthFeb         6.845     4.287   1.597 0.110398
## mnthMarch      16.551     4.301   3.848 0.000120 ***
## mnthApril      41.425     4.972   8.331 < 2e-16 ***
## mnthMay       72.557     5.641  12.862 < 2e-16 ***
## mnthJune      67.819     6.544  10.364 < 2e-16 ***
## mnthJuly      45.324     7.081   6.401 1.63e-10 ***
## mnthAug       53.243     6.640   8.019 1.21e-15 ***
## mnthSept      66.678     5.925  11.254 < 2e-16 ***
## mnthOct       75.834     4.950  15.319 < 2e-16 ***
## mnthNov       60.310     4.610  13.083 < 2e-16 ***
## mnthDec       46.458     4.271  10.878 < 2e-16 ***
## hr1          -14.579     5.699  -2.558 0.010536 *
## hr2          -21.579     5.733  -3.764 0.000168 ***
## hr3          -31.141     5.778  -5.389 7.26e-08 ***
## hr4          -36.908     5.802  -6.361 2.11e-10 ***
## hr5          -24.135     5.737  -4.207 2.61e-05 ***
## hr6           20.600     5.704   3.612 0.000306 ***
## hr7          120.093     5.693  21.095 < 2e-16 ***
## hr8          223.662     5.690  39.310 < 2e-16 ***
## hr9          120.582     5.693  21.182 < 2e-16 ***
## hr10          83.801     5.705  14.689 < 2e-16 ***
## hr11         105.423     5.722  18.424 < 2e-16 ***
## hr12         137.284     5.740  23.916 < 2e-16 ***
## hr13         136.036     5.760  23.617 < 2e-16 ***
## hr14         126.636     5.776  21.923 < 2e-16 ***
## hr15         132.087     5.780  22.852 < 2e-16 ***
## hr16         178.521     5.772  30.927 < 2e-16 ***

```

```
## hr17                296.267        5.749  51.537 < 2e-16 ***
## hr18                269.441        5.736  46.976 < 2e-16 ***
## hr19                186.256        5.714  32.596 < 2e-16 ***
## hr20                125.549        5.704  22.012 < 2e-16 ***
## hr21                 87.554        5.693  15.378 < 2e-16 ***
## hr22                 59.123        5.689  10.392 < 2e-16 ***
## hr23                 26.838        5.688   4.719 2.41e-06 ***
## workingday           1.270         1.784   0.711 0.476810
## temp               157.209       10.261  15.321 < 2e-16 ***
## weathersitcloudy/misty -12.890        1.964  -6.562 5.60e-11 ***
## weathersitlight rain/snow -66.494        2.965 -22.425 < 2e-16 ***
## weathersitheavy rain/snow -109.745       76.667  -1.431 0.152341
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 76.5 on 8605 degrees of freedom
## Multiple R-squared:  0.6745, Adjusted R-squared:  0.6731
## F-statistic: 457.3 on 39 and 8605 DF,  p-value: < 2.2e-16
```

```
anova(modp,test="Chi")
```

```
## Analysis of Variance Table
##
## Response: bikers
##           Df    Sum Sq Mean Sq F value Pr(>F)
## mnth       11 18881416 1716492 293.2733 <2e-16 ***
## hr         23 80952290 3519665 601.3563 <2e-16 ***
## workingday  1      156      156  0.0266 0.8705
## temp        1 1568096 1568096 267.9189 <2e-16 ***
## weathersit   3 2977763 992588 169.5897 <2e-16 ***
## Residuals 8605 50364008    5853
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
drop1(modp,test="Chi")
```

```
## Single term deletions
##
## Model:
## bikers ~ mnth + hr + workingday + temp + weathersit
##           Df Sum of Sq    RSS   AIC Pr(>Chi)
## <none>                 50364008 75033
## mnth       11  3243161 53607169 75550  <2e-16 ***
## hr         23 64684944 115048951 82128  <2e-16 ***
## workingday  1     2963  50366970 75031   0.4758
## temp        1  1373813  51737821 75263  <2e-16 ***
## weathersit   3  2977763  53341771 75523  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

All variables except workingday are significant.

Negative Binomial

```

library(MASS)

modn <- glm( bikers ~ mnth + hr + workingday + temp + weathersit,negative.binomial(1),data = Bikeshare)
summary(modn)

##
## Call:
## glm(formula = bikers ~ mnth + hr + workingday + temp + weathersit,
##      family = negative.binomial(1), data = Bikeshare)
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      2.72052    0.03609   75.379 < 2e-16 ***
## mnthFeb           0.20796    0.02931    7.096 1.38e-12 ***
## mnthMarch         0.31191    0.02939   10.613 < 2e-16 ***
## mnthApril         0.61179    0.03388   18.057 < 2e-16 ***
## mnthMay           0.88510    0.03837   23.069 < 2e-16 ***
## mnthJune          0.84713    0.04446   19.056 < 2e-16 ***
## mnthJuly          0.76024    0.04809   15.809 < 2e-16 ***
## mnthAug           0.77704    0.04510   17.229 < 2e-16 ***
## mnthSept          0.87605    0.04026   21.759 < 2e-16 ***
## mnthOct           0.88273    0.03368   26.209 < 2e-16 ***
## mnthNov           0.78614    0.03139   25.048 < 2e-16 ***
## mnthDec           0.68513    0.02910   23.541 < 2e-16 ***
## hr1              -0.48094    0.03883  -12.385 < 2e-16 ***
## hr2              -0.81400    0.03925  -20.739 < 2e-16 ***
## hr3              -1.43956    0.04009  -35.908 < 2e-16 ***
## hr4              -2.01717    0.04113  -49.042 < 2e-16 ***
## hr5              -0.97513    0.03939  -24.756 < 2e-16 ***
## hr6               0.42264    0.03860   10.950 < 2e-16 ***
## hr7               1.42735    0.03840   37.169 < 2e-16 ***
## hr8               1.99035    0.03835   51.901 < 2e-16 ***
## hr9               1.49135    0.03839   38.844 < 2e-16 ***
## hr10              1.13937    0.03850   29.591 < 2e-16 ***
## hr11              1.27892    0.03861   33.128 < 2e-16 ***
## hr12              1.47444    0.03872   38.084 < 2e-16 ***
## hr13              1.47010    0.03885   37.839 < 2e-16 ***
## hr14              1.42116    0.03896   36.473 < 2e-16 ***
## hr15              1.44892    0.03899   37.163 < 2e-16 ***
## hr16              1.66686    0.03892   42.826 < 2e-16 ***
## hr17              2.12649    0.03874   54.886 < 2e-16 ***
## hr18              2.04179    0.03866   52.819 < 2e-16 ***
## hr19              1.72932    0.03852   44.891 < 2e-16 ***
## hr20              1.42030    0.03847   36.918 < 2e-16 ***
## hr21              1.15880    0.03842   30.160 < 2e-16 ***
## hr22              0.91429    0.03842   23.797 < 2e-16 ***
## hr23              0.52196    0.03847   13.570 < 2e-16 ***
## workingday       -0.20004    0.01208  -16.564 < 2e-16 ***
## temp              0.95742    0.06951   13.773 < 2e-16 ***
## weathersitcloudy/misty -0.06350    0.01329  -4.776 1.81e-06 ***
## weathersitlight rain/snow -0.56524    0.02022 -27.950 < 2e-16 ***
## weathersitheavy rain/snow -0.81445    0.51938  -1.568 0.117
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```
##
## (Dispersion parameter for Negative Binomial(1) family taken to be 0.2613541)
##
## Null deviance: 11404.1 on 8644 degrees of freedom
## Residual deviance: 2735.1 on 8605 degrees of freedom
## AIC: 94664
##
## Number of Fisher Scoring iterations: 10
anova(modn,test="Chi")

## Analysis of Deviance Table
##
## Model: Negative Binomial(1), link: log
##
## Response: bikers
##
## Terms added sequentially (first to last)
##
##
## Df Deviance Resid. Df Resid. Dev Pr(>Chi)
## NULL 8644 11404.1
## mnth 11 1171.0 8633 10233.1 < 2.2e-16 ***
## hr 23 7191.6 8610 3041.5 < 2.2e-16 ***
## workingday 1 71.3 8609 2970.2 < 2.2e-16 ***
## temp 1 54.8 8608 2915.4 < 2.2e-16 ***
## weathersit 3 180.3 8605 2735.1 < 2.2e-16 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
drop1(modn,test="Chi")

## Single term deletions
##
## Model:
## bikers ~ mnth + hr + workingday + temp + weathersit
## Df Deviance AIC scaled dev. Pr(>Chi)
## <none> 2735.1 94664
## mnth 11 3076.6 95949 1306.3 < 2.2e-16 ***
## hr 23 9134.1 119102 24484.0 < 2.2e-16 ***
## workingday 1 2798.4 94904 241.8 < 2.2e-16 ***
## temp 1 2783.7 94848 185.7 < 2.2e-16 ***
## weathersit 3 2915.4 95348 689.7 < 2.2e-16 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
pred.dat<-data.frame(cbind(Y=Bikeshare$bikers,Yhat_P=modp$fitted.values,Yhat_NP=modn$fitted.values))
sum((pred.dat$Y-pred.dat$Yhat_P)^2)

## [1] 50364008
sum((pred.dat$Y-pred.dat$Yhat_NP)^2)

## [1] 48903102
All variables are significant.
```